

CY1010 Environmental Chemistry

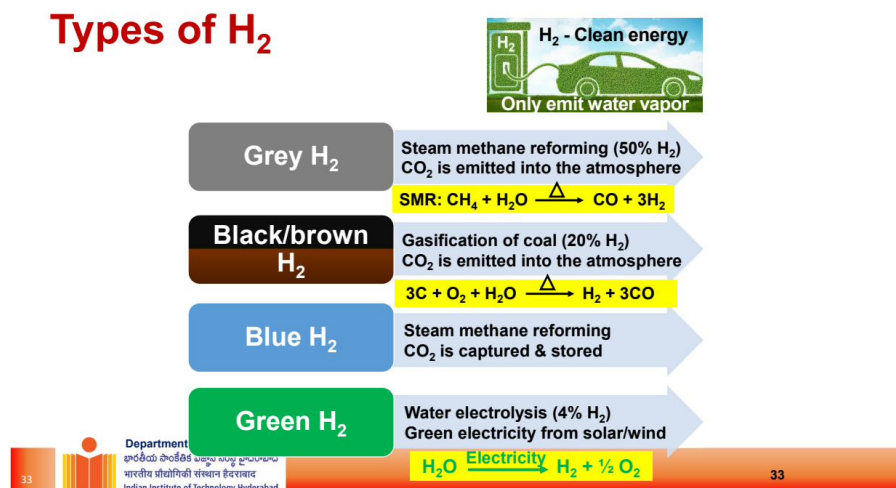
Visual Reactions & Molecular Structures Companion

Exam 1 syllabus — Dr. Anup Bhunia & Dr. Abhijit Sau portions (Lectures 1–9)

This companion collects every key reaction scheme and molecular structure discussed in lecture, drawn out visually rather than described in words — because reaction mechanisms and structures are learned far better by seeing them. Pair this with the main text study guide.

Part 1 — Energy & Catalysis Reactions

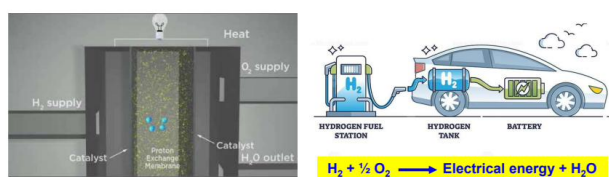
1. The Four Types of Hydrogen (H₂) — full production map



Grey H₂ (SMR, $\text{CH}_4 + \text{H}_2\text{O} \rightarrow \text{CO} + 3\text{H}_2$) and Black/Brown H₂ (coal gasification, $3\text{C} + \text{O}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2 + 3\text{CO}$) both release CO₂. Blue H₂ = SMR with carbon capture. Green H₂ = water electrolysis ($\text{H}_2\text{O} \rightarrow \text{H}_2 + \frac{1}{2}\text{O}_2$) powered by renewables. Memorize this diagram completely — it is a favorite MCQ source.

2. Hydrogen Fuel Cell (electric vehicles)

Hydrogen fuel cell electric vehicles

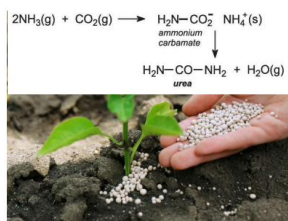


Overall: $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{Electrical energy} + \text{H}_2\text{O}$. Anode: Pt catalyst splits H₂ into protons + electrons. Cathode: Pt catalyst enables O₂ reduction with those protons to form water.

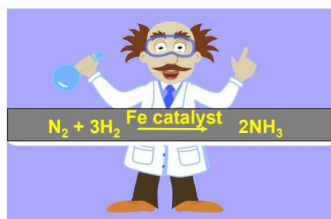
3. Haber–Bosch Process (ammonia synthesis) — most energy-intensive industrial process (1.5–2% of world energy)

Why is catalyst vital?

What is the most energy-intensive process/technology in the world?



Fertilizers
(urea, ammonium sulphate,
ammonium phosphate)



Ammonia synthesis
(Haber-Bosch process)

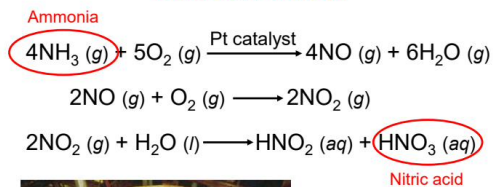
Ammonia synthesis is the most energy-intensive process in the world

- 1.5%–2% of global energy consumption

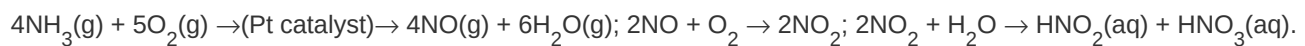
$\text{N}_2 + 3\text{H}_2 \xrightarrow{(\text{Fe catalyst})} 2\text{NH}_3$. Used to make fertilizers (urea, ammonium sulphate, ammonium phosphate). Urea itself:
 $2\text{NH}_3(\text{g}) + \text{CO}_2(\text{g}) \rightarrow \text{ammonium carbamate} \rightarrow (\text{dehydration}) \rightarrow \text{urea} + \text{H}_2\text{O}$.

4. Ostwald Process (nitric acid production)

Ostwald Process

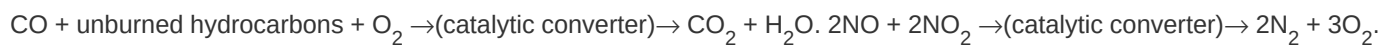
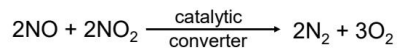
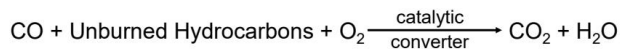
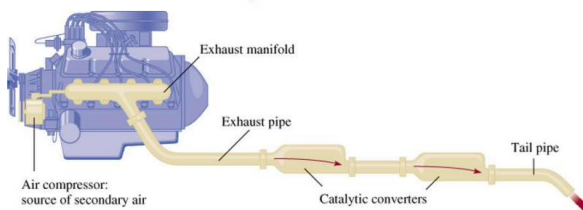


Pt-Rh catalysts used in Ostwald process



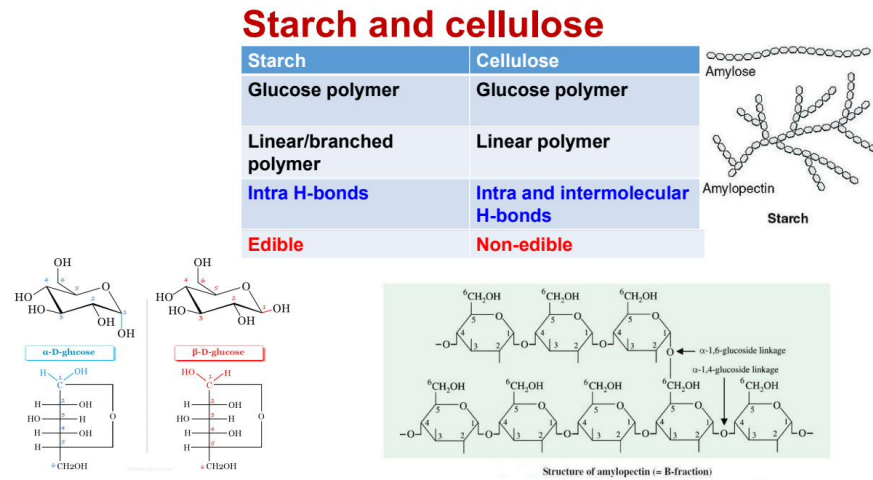
5. Catalytic Converters (automobile exhaust)

Catalytic Converters



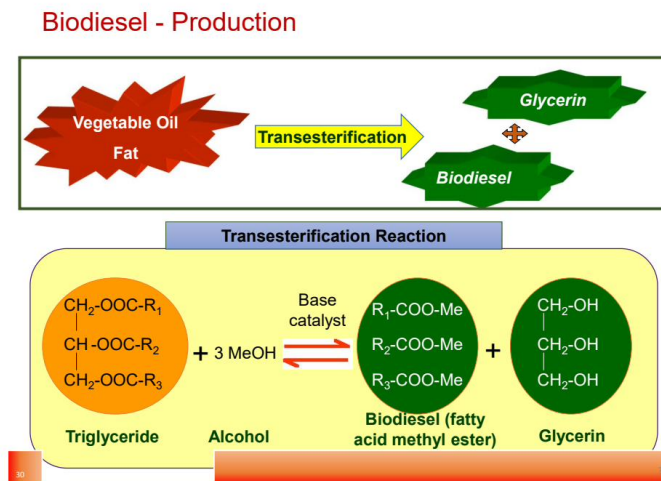
Part 2 — Biomass & Biofuel Structures

6. Starch vs Cellulose — structures and H-bonding (note: reversed from intuition!)



Both are glucose polymers. Starch (amylose linear + amylopectin branched) has ONLY intramolecular H-bonds → edible.
 Cellulose is a linear polymer with BOTH intra- and intermolecular H-bonds (making it rigid/crystalline) → non-edible.
 α-D-glucose vs β-D-glucose anomers shown bottom-left.

7. Transesterification — biodiesel production

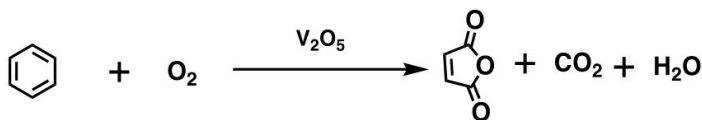


Triglyceride (vegetable oil/fat) + 3 MeOH $\xrightarrow{\text{(base catalyst)}}$ Biodiesel (fatty acid methyl esters, FAME) + Glycerol/Glycerin.

Part 3 — Microbial Degradation & Aromatic Oxidation (Dr. Sau)

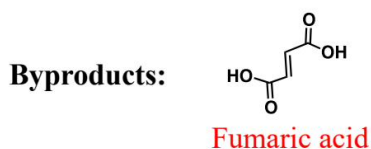
8. Benzene → Maleic Anhydride (chemical oxidation)

Chemical Oxidation of Benzene



Oxidation of benzene produce maleic anhydride

Benzene and air mixture is oxidized to maleic anhydride over the catalyst at 2-5 bar, 400-450 °C,

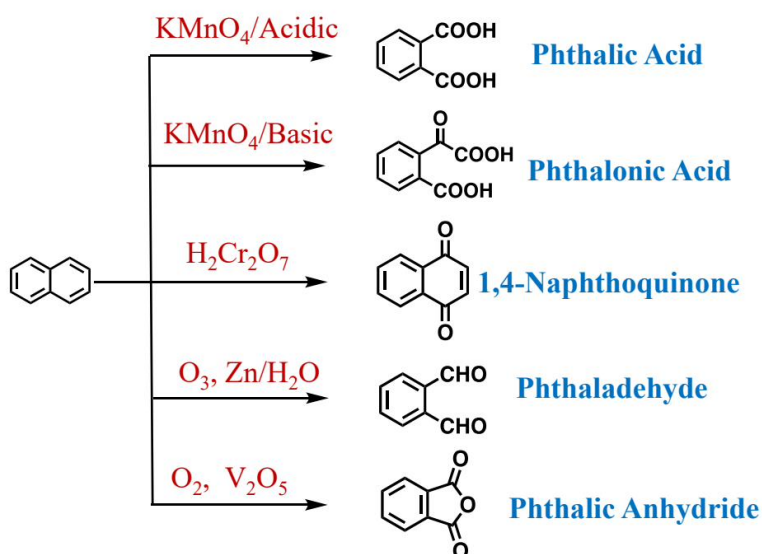


7

Benzene + $O_2 \rightarrow (V_2O_5 \text{ catalyst, 2-5 bar, 400-450}^\circ\text{C}) \rightarrow \text{Maleic anhydride} + CO_2 + H_2O$. Byproduct: Fumaric acid (the trans isomer of the ring-opened diacid).

9. Naphthalene Oxidation Products — all 5 reagent/product pairs (exact matching is commonly tested)

Oxidation Products of Naphthalene



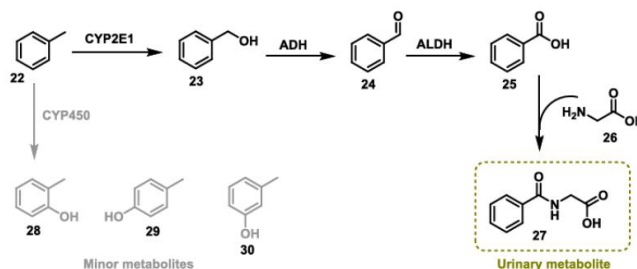
10

$KMnO_4$ /Acidic → Phthalic Acid. $KMnO_4$ /Basic → Phthalonic Acid. $H_2Cr_2O_7 \rightarrow 1,4\text{-Naphthoquinone}$. $O_3, Zn/H_2O \rightarrow \text{Phthalaldehyde}$. $O_2, V_2O_5 \rightarrow \text{Phthalic Anhydride}$.

10. Toluene Metabolism → Hippuric Acid (urinary metabolite)

Oxidation of the cyclic hydrocarbons and aromatic compounds

Scheme 3. Toluene Metabolism



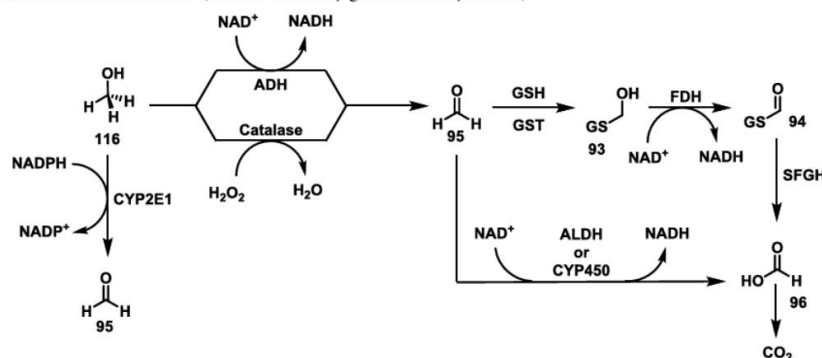
Hippuric acid is biosynthesized from benzoic acid and glycine in the liver, intestine, and kidneys, with benzoic acid first converted to benzoyl-CoA, an activated acylating agent that subsequently reacts with glycine to form hippuric acid.

[ACS Chem.Health Saf.](https://doi.org/10.1021/acs.chas.6c00038)
<https://doi.org/10.1021/acs.chas.6c00038>

Toluene →(CYP2E1)→ benzyl alcohol →(ADH)→ benzaldehyde →(ALDH)→ benzoic acid; benzoic acid + glycine → Hippuric acid (via benzoyl-CoA activation) — the main urinary detox metabolite. Minor CYP450 metabolites: cresols (o-, p-, m-).

11. Methanol Metabolism → Formaldehyde (toxic first metabolite)

Scheme 16. Methanol Metabolism (SFGH = S-Formylglutathione Hydrolase)



The first metabolite formed during methanol oxidation is formaldehyde (95), which is responsible for neuronal damage in the brain and other toxic effects.

[ACS Chem.Health Saf.](https://doi.org/10.1021/acs.chas.6c00038)
<https://doi.org/10.1021/acs.chas.6c00038>

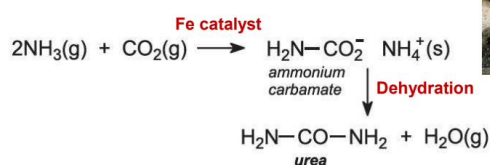
Methanol →(ADH/Catalase)→ Formaldehyde (95) — responsible for neuronal damage — then further oxidized (via glutathione conjugation/FDH/SFGH or ALDH/CYP450) → formic acid → CO₂.

Part 4 — CO₂ Utilization & Green Chemistry Reactions (Dr. Sau)

12. Urea Synthesis from CO₂

Urea synthesis using CO₂

- Urea is synthesized from CO₂ and NH₃.
- The reaction generally operates at 150-250 °C and at a pressure of 2 MPa.
- It is mainly used in the production of fertilizers.

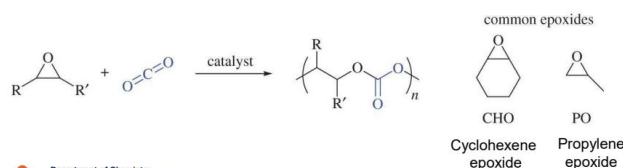


$2\text{NH}_3(\text{g}) + \text{CO}_2(\text{g}) \xrightarrow{(150-250^\circ\text{C}, 2\text{ MPa})} \text{ammonium carbamate} \xrightarrow{\text{dehydration}} \text{Urea} + \text{H}_2\text{O}$. Largest single CO₂-consuming chemical process (~130 Mt CO₂/yr across urea+salicylic acid+polycarbonates).

13. Polycarbonates from CO₂ (ring-opening copolymerization)

Polycarbonates synthesis using CO₂

- Polycarbonates are synthesized via ring-opening copolymerization of epoxides with carbon dioxide.
- ZnEt₂ is a catalyst.
- Examples: plastic lenses in eyewear, automotive components, protective gear, greenhouses, digital Disks (CDs, DVDs, and Blu-ray).

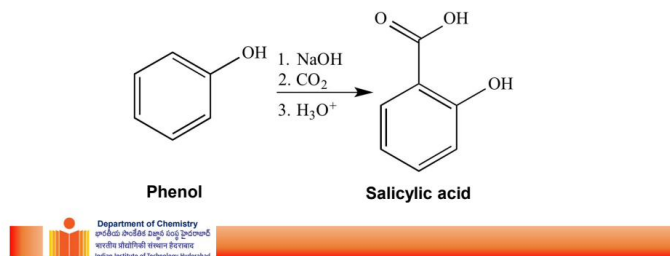


Epoxide + CO₂ $\xrightarrow{(\text{ZnEt}_2 \text{ catalyst})}$ Polycarbonate. Common epoxides: cyclohexene oxide (CHO), propylene oxide (PO).
Uses: eyewear lenses, auto parts, CDs/DVDs/Blu-ray.

14. Salicylic Acid Synthesis — Kolbe-Schmitt Process

Salicylic acid synthesis using CO₂

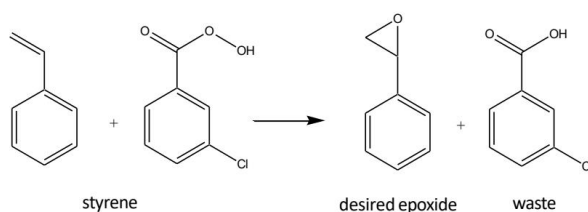
- Salicylic acid is used **to treat skin disorders**.
- It is produced commercially **via the Kolbe-Schmitt process**.
- Phenol and sodium hydroxide are reacted to form sodium phenoxide. Then phenoxide reacts with CO₂ to form sodium salicylate. The salicylate is acidified to give salicylic acid.



Phenol $\rightarrow$ (1. NaOH, 2. CO₂, 3. H₃O⁺) $\rightarrow$ Salicylic acid. Phenol + NaOH $\rightarrow$ sodium phenoxide; phenoxide + CO₂ $\rightarrow$ sodium salicylate; acidify $\rightarrow$ salicylic acid.

15. Atom Economy Case Study — Styrene Epoxidation (23% atom economy)

Example: Epoxidation of styrene



- Assume 100% yield.
- 100% of the desired epoxide product is recovered.
- 100% formation of the co-product: m-chlorobenzoic acid.
- A.E. of this reaction is 23%.
- 77% of the products are waste.

Styrene + m-CPBA (a peroxyacid) $\rightarrow$ desired epoxide + m-chlorobenzoic acid (waste, 77% of mass). Classic example of a LOW atom-economy oxidation reaction.

16. Atom Economy Case Study — Ethylene Oxide: Old (Chlorine) Route

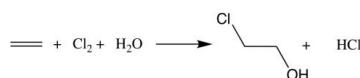
Case Study: Production of Ethylene Oxide

Ethylene oxide is used as an intermediate in the production of several industrial chemicals, the most notable of which is ethylene glycol. It is also used as a fumigant in certain agricultural products and as a sterilant for medical equipment and supplies.

Conventional ethylene oxide synthesis included:

- A 2-step synthesis with a chlorohydrin intermediate.
- For each kilogram of product, 5 Kg of waste were disposed.

Step 1:



Step 2:



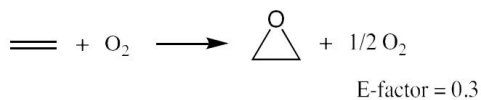
Step 1: Ethylene + Cl_2 + H_2O $\rightarrow$ chlorohydrin + HCl. Step 2: Chlorohydrin + Ca(OH)_2 $\rightarrow$ Ethylene oxide + CaCl_2 + H_2O .
Conventional route: 5 kg waste generated per kg of product.

17. Atom Economy Case Study — Ethylene Oxide: Greener (O_2) Route

Case Study: Production of Ethylene Oxide

Alternative production of ethylene oxide:

- Use of molecular oxygen removes the need for chlorine.
- New process generates more than 16 times less waste than the original one, and eliminates the formation of waste water.



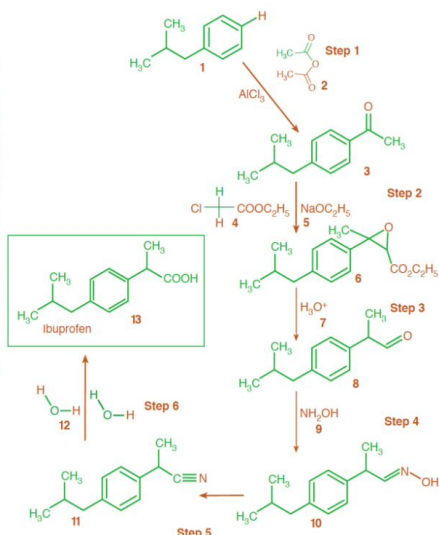
Ethylene + O_2 $\rightarrow$ Ethylene oxide + $\frac{1}{2}\text{O}_2$ (catalytic, no chlorine needed). E-factor = 0.3 — over 16× less waste than the chlorine route, and eliminates wastewater formation entirely.

18. Ibuprofen Synthesis — Boots Process (original, 6 steps, poor atom economy)

Atom economy example: Ibuprofen synthesis

The Problem:

The traditional industrial synthesis of ibuprofen was developed and patented by the Boots Company of England in the 1960s (U.S. Patent 3,385,886). This synthesis is a **six-step process** and **results in large quantities of unwanted waste** chemical byproducts that must be disposed of or otherwise managed. Much of the waste that is generated is a result of **many of the atoms of the reactants not being incorporated into the desired product** (ibuprofen) but into unwanted byproducts (poor atom economy/atom utilization). The process also uses a variety of solvents, and catalysts (if any) are not used in stoichiometric amounts.



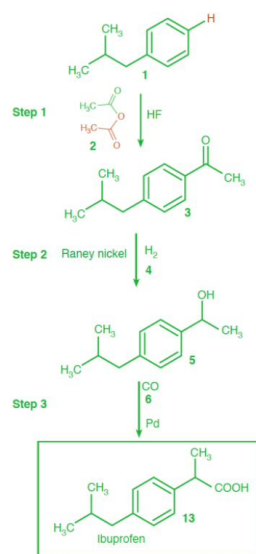
Uses stoichiometric AlCl_3 , NaOC_2H_5 , NH_2OH and others; generates large quantities of unwanted byproducts because many atoms of the reactants are never incorporated into the final ibuprofen product.

19. Ibuprofen Synthesis — BHC Process (greener, 1991, only 3 steps)

Atom economy example: Ibuprofen synthesis

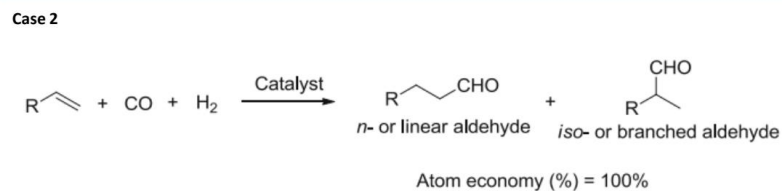
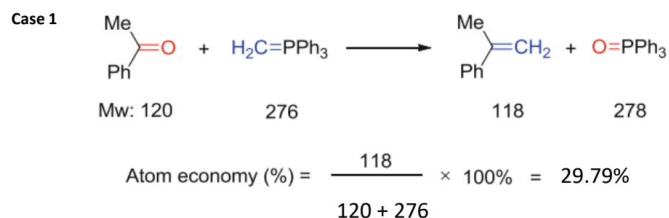
The Solution:

The BHC Company has developed and implemented a new greener industrial **synthesis of ibuprofen that is only three steps** (U.S. Patents 4,981,995 and 5,068,448, both issued in 1991). In this process, **most of the atoms of the reactants are incorporated into the desired product** (ibuprofen). This results in only small amounts of unwanted byproducts (very good atom economy/atom utilization) thus lessening the need for disposal and mediation of waste products. There are other environmental advantages to the green synthesis versus the brown synthesis, including solvents and catalyst.



Step 1: HF-catalyzed Friedel-Crafts acylation. Step 2: H₂ / Raney Nickel reduction. Step 3: CO / Pd carbonylation. Most atoms of the reactants end up in the final product — very good atom economy, and HF is recovered/recycled.

20. Atom Economy Contrast — Wittig Reaction (29.8%) vs Hydroformylation (100%)

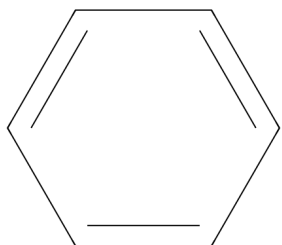


Case 1 (Wittig, a substitution-type reaction): $\text{PhC(Me)=O} + \text{Ph}_3\text{P=CH}_2 \rightarrow \text{alkene (desired, MW 118)} + \text{Ph}_3\text{P=O (waste, MW 278)}$. Atom economy = $118/(120+276) \times 100 = 29.79\%$. Case 2 (Hydroformylation, an addition reaction): $\text{alkene} + \text{CO} + \text{H}_2 \xrightarrow{\text{catalyst}} \text{aldehyde}$. Atom economy = 100% because every atom of every reactant ends up in the product. Key lesson: addition reactions are inherently atom-economical; substitution/elimination reactions generate stoichiometric waste.

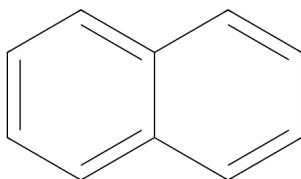
Part 5 — Organic Molecule Structure Gallery

Every organic molecule named in Lectures 1–9, drawn out for quick visual recognition. Grouped by theme; use this to test yourself by covering the name and identifying the structure.

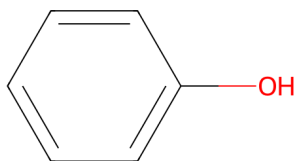
Aromatic hydrocarbons & simple derivatives



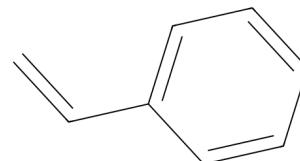
Benzene



Naphthalene

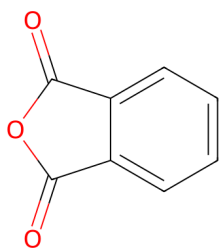


Phenol

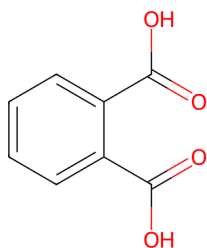


Styrene

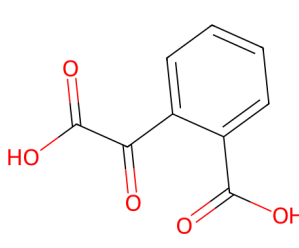
Naphthalene oxidation products



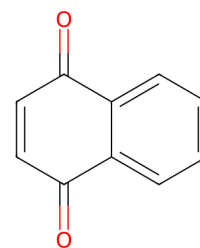
Phthalic Anhydride



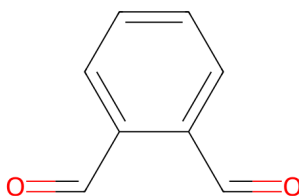
Phthalic Acid



Phthalonic Acid

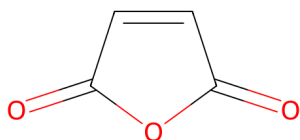


1,4-Naphthoquinone

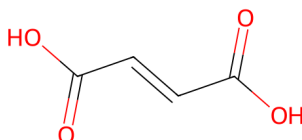


Phthalaldehyde

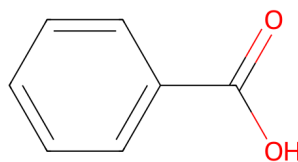
Benzene oxidation & related acids



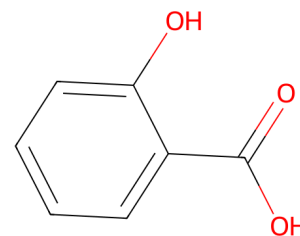
Maleic Anhydride



Fumaric Acid



Benzoic Acid

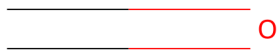


Salicylic Acid

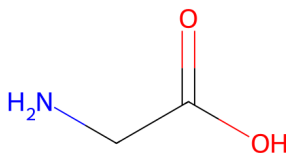
Metabolism intermediates



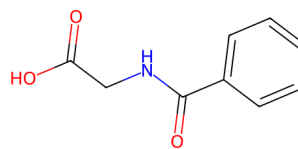
Methanol



Formaldehyde

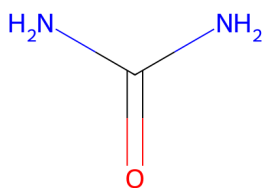


Glycine

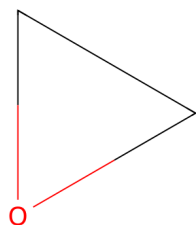


Hippuric Acid

Green chemistry & CO2-utilization products



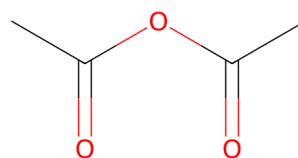
Urea



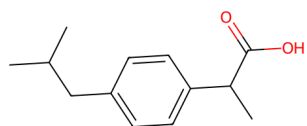
Ethylene Oxide



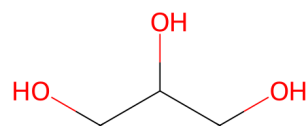
Ethylene Glycol



Acetic Anhydride

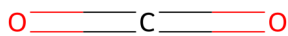


Ibuprofen



Glycerol (from biodiesel)

Key gas molecules (greenhouse gases & pollutants)



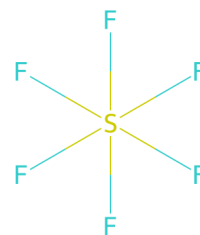
Carbon Dioxide (CO₂)



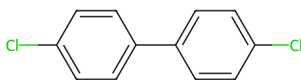
Methane (CH₄)



Nitrous Oxide (N₂O)



Sulfur Hexafluoride (SF₆) —
highest GWP



PCB (example POP structure)

Part 6 — GHS Hazard Pictograms & Other Key Diagrams

21. GHS Physical Hazard Pictograms

Physical Hazard Pictograms



Black symbol, white background, red diamond frame. Shown: Explosives, Corrosion, Gas Under Pressure, Oxidizers (flame over circle), Flammables.

22. GHS Health Hazard Pictograms

Health Hazard Pictograms



Health Hazard (carcinogen/mutagen/respiratory sensitizer/target organ toxicity/aspiration toxicity), Exclamation Mark (irritant), Skull & Crossbones (acute toxicity), Corrosion (skin/eye damage).

23. GHS Environmental Hazard Pictogram

Environment



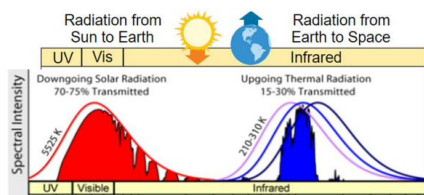
Aquatic toxicity

Dead tree + fish = aquatic toxicity / hazardous to the environment. Together with the previous two slides, this completes all 9 standard GHS pictograms.

24. Atmospheric Window

Atmospheric window

An **atmospheric window** is the portion of the electromagnetic spectrum that can be transmitted through the atmosphere.



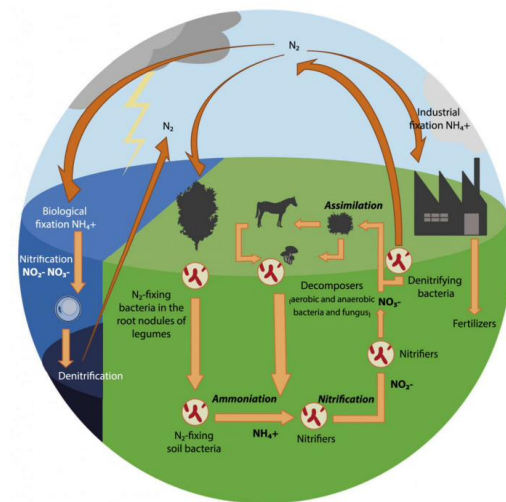
Earth's Energy Balance:

The infrared atmospheric window plays a crucial role in Earth's energy balance by allowing thermal radiation from the surface to escape into space, which helps regulate the planet's temperature.

- The filled-in red area is the spectrum of the sunlight that reaches the earth and the missing radiation (25-30%) is either absorbed by GHGs or scattered by O_2/N_2 molecules.
- The filled-in blue area is the spectrum of the sunlight that released/reflected from earth to space and the remaining radiation (70-85%) is absorbed by GHGs.

70–75% of incoming solar (UV/Vis) radiation is transmitted to Earth; the missing 25–30% is absorbed by GHGs or scattered by O_2/N_2 . Only 15–30% of outgoing infrared (thermal) radiation escapes to space — the rest is absorbed by GHGs, driving the greenhouse effect.

25. Nitrogen Cycle



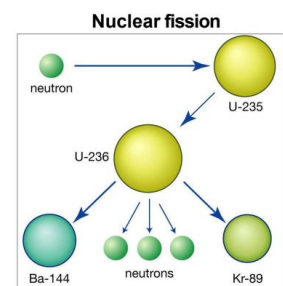
One critical example is nitrogen, which makes up 78% of the atmosphere as nitrogen gas, but which is not usable in that form by most living things. Nitrogen cycles via the processes of nitrogen fixation, ammonification, nitrification, assimilation, decomposition, and denitrification.

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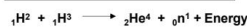
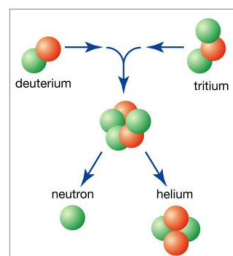
N_2 (78% of atmosphere, unusable directly) cycles via: nitrogen fixation (biological/industrial) → ammonification → nitrification → assimilation → decomposition → denitrification (returns N_2 to atmosphere).

26. Nuclear Fission vs Nuclear Fusion

Nuclear energy



Nuclear fusion

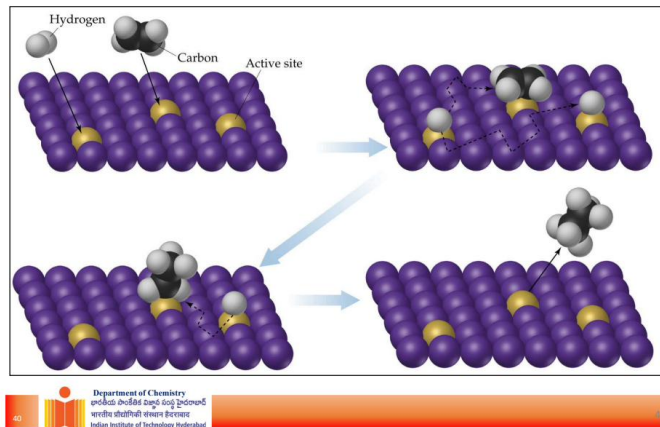


Four nuclear power stations in India: Tarapur (Maharashtra), Rana Pratap Sagar (Rajasthan), Kalpakkam (Tamilnadu), and Narora (U.P)

Fission: $U-235 + \text{neutron} \rightarrow U-236 \rightarrow Ba-144 + Kr-89 + 3 \text{ neutrons} + \text{Energy}$. Fusion: Deuterium (${}_1H^2$) + Tritium (${}_1H^3$) → Helium (${}_2He^4$) + neutron + Energy.

27. Heterogeneous Catalysis Mechanism (surface catalysis, step-by-step)

Heterogeneous Catalysts



Example: $\text{C}_2\text{H}_4(\text{g}) + \text{H}_2(\text{g}) \rightarrow (\text{metal surface}) \rightarrow \text{C}_2\text{H}_6(\text{g})$. Steps shown: (1) adsorption of both reactants onto active sites, (2) diffusion/bond activation on the surface, (3) surface reaction forming the product, (4) desorption of product from the surface.

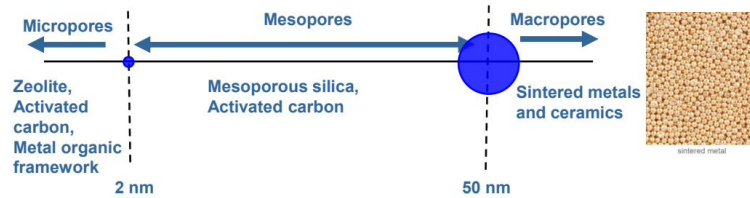
28. Physisorption vs Chemisorption

PHYSISORPTION	CHEMISORPTION
WEAK, LONG-RANGE BONDING van der Waals interactions	STRONG, SHORT-RANGE BONDING Chemical bonding involved.
NOT SURFACE SPECIFIC Physisorption takes place between all molecules on any surface providing the temperature is low enough.	SURFACE SPECIFIC E.g. Chemisorption of hydrogen takes place on transition metals but not on gold or mercury.
$\Delta H_{\text{ads}} = 5 \dots 50 \text{ kJ mol}^{-1}$	$\Delta H_{\text{ads}} = 50 \dots 500 \text{ kJ mol}^{-1}$
Non activated with equilibrium achieved relatively quickly. Increasing temperature always reduces surface coverage.	Can be activated, in which case equilibrium can be slow and increasing temperature can favour adsorption.
No surface reactions.	Surface reactions may take place:- Dissociation, reconstruction, catalysis.

Physisorption: weak van der Waals bonding, not surface-specific, $\Delta H = 5\text{--}50 \text{ kJ/mol}$, higher T reduces coverage, no surface reactions. Chemisorption: strong chemical bonding, surface-specific, $\Delta H = 50\text{--}500 \text{ kJ/mol}$, can be activated, surface reactions (dissociation/reconstruction/catalysis) may occur.

29. Pore Size Classification (IUPAC)

Size of Pores (IUPAC Standard)



Porous materials are classified according to the size of pores: material with pores less than 2 nm are called micropores, materials with pores between 2 and 50 nm are called mesopores, and material with pores greater than 50 nm are macropores

Sing, K. S. W. et al. *Reporting Physisorption Data for Gas/Solid Systems*. *Pure & Appl. Chem.* 57, 603-619 (1985).



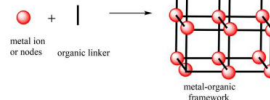
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భారతీయ ఇంజనీరింగ్ సంస్థ
Indian Institute of Technology Hyderabad

Micropores <2 nm (zeolite, activated carbon, MOF) | Mesopores 2–50 nm (mesoporous silica, activated carbon) |
Macropores >50 nm (sintered metals/ceramics).

30. Metal-Organic Frameworks (MOFs)

Metal-Organic Frameworks (MOFs)

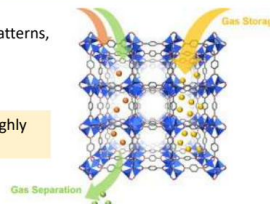
MOFs are porous materials with up to 94% empty space in their framework.



Metal-Organic Frameworks (MOFs) are advanced porous materials with ultra-high surface areas and tunable pore sizes that make them highly effective for gas storage and separation.

Building blocks self-assemble into repeating, grid-like (reticular) patterns, creating structures with tiny, interconnected cavities or pores.

The size and chemical nature of the pores can be designed to be highly specific, allowing MOFs to selectively adsorb certain molecules.

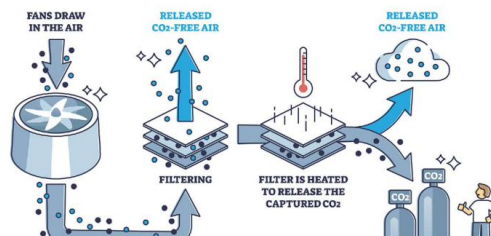


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Indian Institute of Technology Hyderabad

Metal ion/node + organic linker self-assemble into a repeating grid framework with up to 94% empty space — used for gas storage and gas separation. 2025 Nobel Prize in Chemistry (Kitagawa, Robson, Yaghi).

31. Direct Air Capture (DAC) — how it physically works

Direct air capture (DAC)



- **Direct air capture (DAC)** technology uses big machines which contains fan and filters to capture CO_2 from the air.
- When air passes through these machines, **the chemicals/materials inside the filters react with CO_2** , in order to capture it selectively and allow the rest of the air to pass through.
 - Two types of capturing agents:
 1. **Liquid solvents**
 2. **Solid adsorbents**



Fans draw in ambient air → filters (solid sorbents or liquid solvents) selectively react with/capture CO_2 , releasing CO_2 -free air → filter is heated to release the captured CO_2 → concentrated CO_2 collected for storage or utilization.

32. UN Sustainable Development Goals (17 goals)

POPULATION AND THE SUSTAINABLE DEVELOPMENT GOALS: The UN 17 Sustainable Goals

Aim: shared blueprint for peace and prosperity for people and the planet, now and into the future

“Development that meets the needs of the present Without compromising the ability of the future generations to meet their own needs”

poverty, inequality, climate change, environmental degradation, peace and justice

Global partnership among developing and developed countries



<https://sdgs.un.org/goals>

Global blueprint for peace and prosperity for people and the planet. Relevant to green chemistry / industrial ecology discussions of sustainability.